

RESEARCH ARTICLE

Lianas and trees exhibit divergent intrinsic water-use efficiency along elevational gradients in South American and African tropical forests

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Abstract

Aim: Elevational gradients provide excellent opportunities to explore long-term morphological and physiological responses of plants to environmental change. We determined the difference in the elevational pattern of foliar carbon isotope composition ($\delta^{13}\text{C}$) between lianas and trees, and assessed whether this difference arises from changes in photosynthesis or stomatal conductance. We also explored the pattern of nutrient limitations with the elevation of these two growth forms.

Location: The study was conducted in two mountain forests situated in the Neotropics and Palaeotropics.

Time period: August–September 2015 and August–October 2016.

Major taxa studied: Lianas and trees.

Methods: We conducted inventories of lianas and trees using standardized techniques along elevational gradients in Ecuador and Rwanda. We determined the values of several foliar traits including $\delta^{13}\text{C}$ and chemical traits in dominant liana and tree species. We set up Bayesian linear mixed-effect models to quantify the effects of elevation and growth form on each of the foliar traits, and the difference of the effect of elevation between the two growth forms (lianas and trees).

Results: We found consistent growth form specific divergences in foliar $\delta^{13}\text{C}$ and carbon to nitrogen ratio (C : N) responses to elevation. While we noted a meaningful increase in foliar $\delta^{13}\text{C}$ and C : N with elevation for trees, lianas did not exhibit such a trend. Foliar $\delta^{13}\text{C}$ and C : N remained relatively constant for lianas along the transects.

Main conclusions: Lianas operate at relatively constant intrinsic water- and nitrogen-use efficiencies with elevation compared with trees. Altogether, the study suggests the existence of a functional divergence of water and nutrient use strategies between lianas and trees along tropical elevational transects.

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1 | INTRODUCTION

Lianas, woody vines and structural parasites of trees, are reported to be increasing in biomass and abundance relative to trees in Neotropical forests (Schnitzer, 2015). As this relative increase is expected to lead to lower tree growth and higher tree mortality (Estrada-Villegas & Schnitzer, 2018), lianas can severely impact the carbon cycling, storage, and sequestration of tropical forests (van der Heijden et al., 2013), and further exacerbate climatic change (van der Heijden et al., 2015). Alterations in environmental conditions characterized by elevated CO_2 concentrations (Granados & Körner, 2002) and increasing seasonal drought (Schnitzer & Bongers, 2011) are among the probable causes of the observed liana proliferation pattern. Particularly, lianas are known to benefit from a dry seasonal growth advantage over trees (Schnitzer & van der Heijden, 2019). The underlying mechanisms behind this dry seasonal growth advantage of lianas remain an unresolved issue. However, several studies suggest that lianas have better access to water and nutrients, experience less drought stress (Chen et al., 2015; De Deurwaerder et al., 2018), fix more carbon, use water and nitrogen more efficiently (Cai et al., 2009), and have a greater osmotic adjustment (Maréchaux et al., 2017) compared to co-occurring trees during the dry season relative to the wet season.

The investigation of foliar eco-physiological and morphological traits along elevational gradients may allow patterns of resource use between lianas and trees to be distinguished, and assist in evaluating the mechanisms of their growth and survival under changing environmental conditions. Elevational gradients, in particular, are open-air laboratories that can serve as natural experiments to explore ecosystem responses to long-term environmental changes (Malhi et al., 2010). They are characterized by substantial shifts in atmospheric pressure, temperature, and other covarying abiotic factors such as irradiance, precipitation, soil moisture and nutrient availability over short distances (Malhi et al., 2010).

The carbon isotope composition ($\delta^{13}\text{C}$) in C_3 plant tissues is an eco-physiological trait that offers an integrated measure of the ratio between CO_2 partial pressure inside the stomatal cavity and that in the ambient air (c_i/c_a). Hence, it can be used as a surrogate to estimate the ratio of carbon acquired during net assimilation (A_n) to the potential for water vapour losses through stomatal conductance (g_s), that is, intrinsic water use efficiency (iWUE) (Farquhar & Richards, 1984). However, $\delta^{13}\text{C}$ cannot be used to distinguish whether variations in iWUE originate from the independent effects of A_n or g_s . In contrast, the oxygen isotope composition ($\delta^{18}\text{O}$) of plant tissues may be used

to independently quantify variations in iWUE emanating from variations in g_s . When combined with $\delta^{13}\text{C}$, $\delta^{18}\text{O}$ represents an elegant way of separating different sources of variation in iWUE. That is because foliar $\delta^{18}\text{O}$ shares a dependence on g_s with foliar $\delta^{13}\text{C}$, but it is not influenced by variation in A_n (Barbour, 2007; Scheidegger et al., 2000).

The foliar $\delta^{13}\text{C}$ may be associated with other plant morphological and physiological traits such as leaf nitrogen content (LNC), leaf phosphorus content (LPC) and specific leaf area (SLA). For instance, plants with higher nitrogen per unit area (N_{area}) tend to have higher A_n , because photosynthetic enzymes such as Rubisco are rich in N (Walker et al., 2014). In return, higher A_n induces lower ^{13}C - CO_2 discrimination by plants, resulting in higher foliar $\delta^{13}\text{C}$ values (Friend et al., 1989). Similarly, there exists a link between phosphorus per unit area (P_{area}) and A_n via the recycling of the ribulose 1,5-bisphosphate (RuBP) molecule, the transformation of adenosine triphosphate (ATP), and the synthesis of new proteins and nucleic acids (Walker et al., 1986). Furthermore, the relationship between N_{area} and A_n appears to vary with different levels of P_{area} (Reich et al., 2009). Regarding SLA, it may be related to foliar $\delta^{13}\text{C}$ through A_n , but the strength of this relationship depends on the distribution of the mass component of the leaf, which is divided into photosynthetic and structural mass (Osnas et al., 2018).

Besides these leaf traits, foliar elemental ratios—that is, C : N, C : P and N : P—are important plant eco-physiological measures, which reflect carbon and nutrient cycles (de Mello Prado & Pereira da Silva, 2017). The exploration of the changes in these foliar elemental ratios with environmental factors, in connection with foliar $\delta^{13}\text{C}$, may provide a broader view on nutrient and water limitations to primary productivity.

In this study, we measured foliar carbon and oxygen isotopic compositions ($\delta^{13}\text{C}$ and $\delta^{18}\text{O}$), foliar carbon (C), foliar nutrients (N and P) and the specific leaf area (SLA) of the most dominant liana and tree species along two independent elevational gradients in tropical forests located in Ecuador (South America) and Rwanda (East-Central Africa). The aim of the study is threefold: (a) determining the difference in foliar $\delta^{13}\text{C}$ between lianas and trees with elevation, and assessing whether this difference arises from changes in A_n , g_s or both; (b) ascertaining whether nutrient limitation with elevation differs between these two growth forms; and (c) quantifying the extent to which the relative abundance and basal area of lianas change as a function of elevation. We hypothesized that foliar $\delta^{13}\text{C}$ of both growth forms would increase with elevation as a response to the most important growth and survival constraints along these two tropical elevational transects, that is, a decrease in atmospheric pressure and temperature.

However, the absolute increase in $\delta^{13}\text{C}$ would differ between the two growth forms due to differences in internal control mechanisms over A_n or g_s . Because lianas can be very effective in terms of water acquisition and transport, their g_s would be less reduced (relative to A_n) with increasing elevation compared to that of trees. Consequently, c_i would drop less, and $\delta^{13}\text{C}$ would increase less (lower iWUE) for lianas than trees at high elevations. Furthermore, both growth forms would experience a progressive shift towards N limitation with elevation, but this would be less marked for lianas than for trees due to their higher ability for nutrient acquisition. Lastly, the vascular system of lianas would become maladaptive in the cold temperatures associated with higher elevations. Therefore, liana relative abundance and basal area would decrease at higher elevations.

2 | METHODS

2.1 | Study sites, inventories, sampling, and trait analysis

The provinces of Pichincha (0°15' S, 78°35' W) and Imbabura (0°21' N, 78°07' W) in northern Ecuador contain patches of old-growth foothill and montane forests. Located on the western slope of the Andes, they form an elevational gradient ranging from 400–3,200 m a.s.l. in the tropical montane cloud forests. From the lowest to the highest elevation, mean annual temperature ranges from 24.8 to 11.3 °C whereas mean annual precipitation ranges from 3,447 to 1,181 mm. A dry season occurs along the elevational transect between July and November, months during which monthly precipitation is lower than 100 mm (WorldClim—Fick & Hijmans, 2017). Soils across the elevational gradient are classified as Andisols, and the parent materials are of different origins (Table 1).

The Nyungwe Forest Reserve located in south-western Rwanda (2°15'–2°55' S, 29°00'–29°30' E) is part of the Albertine Rift. It constitutes one of the most biologically important montane rain forests in central Africa, comprising large stretches of forests at elevations ranging from 1,600–3,000 m a.s.l. (Plumptre et al., 2002). Temperatures at Nyungwe are generally cool with mean annual temperatures ranging from 19.6 to 10.9 °C along the elevational gradient. Precipitation increases with increasing elevation, and mean annual precipitation is estimated at 1,744 mm. There is a 2-month dry season, occurring between mid-June and mid-August. Soils in the reserve are generally acidic, and strongly varying in soil textural properties (Table 1).

We conducted inventories for lianas (August–October 2016) in 17 and 20 (40 × 40 m) permanent sampling plots (PSPs) established along the Ecuadorian and Rwandan elevational gradients, respectively (Figure 1). These PSPs were spatially clustered around four elevations on both gradients. We used data from tree inventories conducted in the same PSPs from August to September 2015 (Bauters et al., 2017). The liana inventory was based on the protocol of Gerwing et al. (2006), whereas the tree inventory followed an international standardized protocol for tropical forest inventories (Malhi et al., 2002). The diameter of liana stems with a diameter

≥ 2 cm was measured at 1.3-m distance along the stem from the last rooting position while that of tree individuals with a diameter ≥ 10 cm was measured at breast height (1.3 m). We assigned each enumerated stem of both growth forms to a specific species or morphospecies, calculated their basal area (BA) and, whenever possible, identified them to the species or genus level. We then computed the number of species, the total abundance and total BA per plot by summing the values for all individuals present in each plot. The plot-level values of BA were scaled up to per hectare (ha) level. Subsequently, for each of these taxonomic and forest structure attributes, we calculated the relative value (per plot or per ha) of each growth form. We took the ratio of the value of each attribute for each growth form to the total value of the same attribute for both growth forms in each plot (Table 1). We collected leaf samples at the end of each inventory campaign, and determined foliar $\delta^{13}\text{C}$, $\delta^{18}\text{O}$, C, N and P as well as SLA following the extended methods described in Supporting information Appendix S1.

2.2 | Data analysis

To analyse the relationship between elevation, growth form and leaf traits (i.e. $\delta^{13}\text{C}$, $\delta^{18}\text{O}$, N_{area} , P_{area} , C : N, C : P, N : P, SLA), we used Bayesian robust linear mixed-effects models (LMMs). Because plots were spatially clustered around four elevations on both transects, we set these elevational clusters as random intercept effects, with plots nested within elevational clusters as an additional random intercept, whereas elevation and growth form were treated as fixed effects. All LMMs were specified as follows:

$$\mu_{ijk} = a_i + b_{ij} + \beta_i + \beta_E \text{Elevation}_{ijk} + \beta_G \text{Growthform}_{ijk} + \beta_{GE} (\text{Elevation}_{ijk} \times \text{Growthform}_{ijk}) \quad (1)$$

where μ_{ijk} is the mean distribution of the response variable, corresponding to the k th observation at plot j in elevational cluster i , where $i = 1, \dots, 4$, and $j = 1, \dots, p_i$ with p_i the number of plots in elevational cluster i , and $k = 1, \dots, n_j$ with n_j the number of observations at plot j . The term a_i is the random intercept and allows for a random variation on the intercept for the elevational cluster i , and b_{ij} is a random intercept for plot j nested within elevational cluster i , allowing for a different intercept per plot j .

This equation shows that we modelled the response variable (foliar traits) on random intercepts a_i and b_{ij} , and regression coefficients for intercept (β_i), elevation (β_E), growth form (β_G), and the interaction between elevation and growth form (β_{GE}). These parameters captured our questions about the effects of elevation and growth form, and the difference of the effect of elevation between the two growth forms (interaction) on the response variable. The parameters β_i and β_E define, respectively, the effect of the reference growth form group (i.e. the lianas) at elevation 0 m, and the effect of elevation for the reference growth form group on the response variable. Similarly, β_G quantified the effect of the non-reference growth form group (i.e. the trees) on the

TABLE 1 General characteristics, forest structure, climate [mean annual temperature (MAT) and mean annual precipitation (MAP), WorldClim—Fick & Hijmans, 2017] and soil characteristics of elevational clusters on the Ecuadorian and Rwandan transects

Cluster	n	Elevation (m a.s.l.)	Abundance (stems per plot)		Relative abundance (%)		Species richness per plot			Relative species richness (%)		Basal area (m ² /ha)			Relative basal area (%)			MAT (°C)	MAP (mm)	Soil parent material	Soil classification	
			L	T	L	T	L	T	L	T	L	T	L	T								
Ecuador																						
C1	5	407 (9)	35 (12)	83 (16)	30 (11)	70 (11)	7 (2)	29 (2)	20 (5)	80 (5)	0.31 (0.14)	23 (4)	1.4 (0.7)	98.6 (0.7)	23.7	3,720	Lahars	Andisol				
C2	4	1,068 (25)	30 (15)	82 (38)	27 (5)	73 (5)	7 (2)	38 (12)	17 (6)	83 (6)	0.2 (0.1)	27 (8)	0.7 (0.1)	99.3 (0.1)	20.0	3,227	Lahars	Andisol				
C3	4	1,871 (79)	55 (24)	66 (12)	44 (9)	56 (9)	9 (3)	30 (3)	24 (7)	76 (7)	0.36 (0.24)	27 (8)	1.4 (0.9)	98.6 (0.9)	17.5	1,619	Redbed Volcaniclastics	Andisol				
C4	4	3,217 (21)	27 (15)	94 (24)	23 (15)	77 (15)	7 (2)	18 (2)	27 (5)	73 (5)	0.13 (0.1)	52 (14)	0.3 (0.2)	99.7 (0.2)	10.9	1,241	Granitic/ acid	Andisol				
Rwanda																						
C1	5	1,760 (66)	54 (13)	70 (18)	44 (7)	56 (7)	12 (3)	22 (5)	35 (4)	65 (4)	0.58 (0.25)	34 (4)	1.7 (0.7)	98.3 (0.7)	17.6	1518	Shale and quartzite	Inceptisol/ Ultisol				
C2	5	2,200 (64)	30 (23)	70 (18)	27 (17)	73 (17)	7 (4)	19 (3)	26 (11)	74 (11)	0.40 (0.32)	41 (10)	1 (0.8)	99 (0.8)	15.9	1628	Shale and quartzite	Inceptisol/ Ultisol				
C3	5	2,512 (37)	143 (61)	121 (60)	55 (12)	45 (12)	9 (2)	11 (1)	45 (6)	55 (6)	1.27 (0.81)	31 (8)	4 (2)	96 (2)	14.7	1716	Shale and quartzite	Inceptisol/ Ultisol				
C4	5	2,844 (77)	8 (6)	109 (56)	9 (7)	91 (7)	2 (1)	8 (2)	23 (10)	77 (10)	0.04 (0.04)	31 (7)	0.2 (0.1)	99.8 (0.1)	12.9	1835	Shale and quartzite	Inceptisol/ Ultisol				

Note: L = lianas; T = trees.

Elevation as well as stem abundance and species richness, relative stem abundance and species richness (in 40 m x 40 m plots), basal area and relative basal area (per ha) of growth forms are averages per elevational cluster along with standard deviations in parentheses on the plot-level results. The letter n indicates the number of plots per altitudinal cluster. The diameter cut-off for lianas was 2 cm, and that of trees 10 cm.

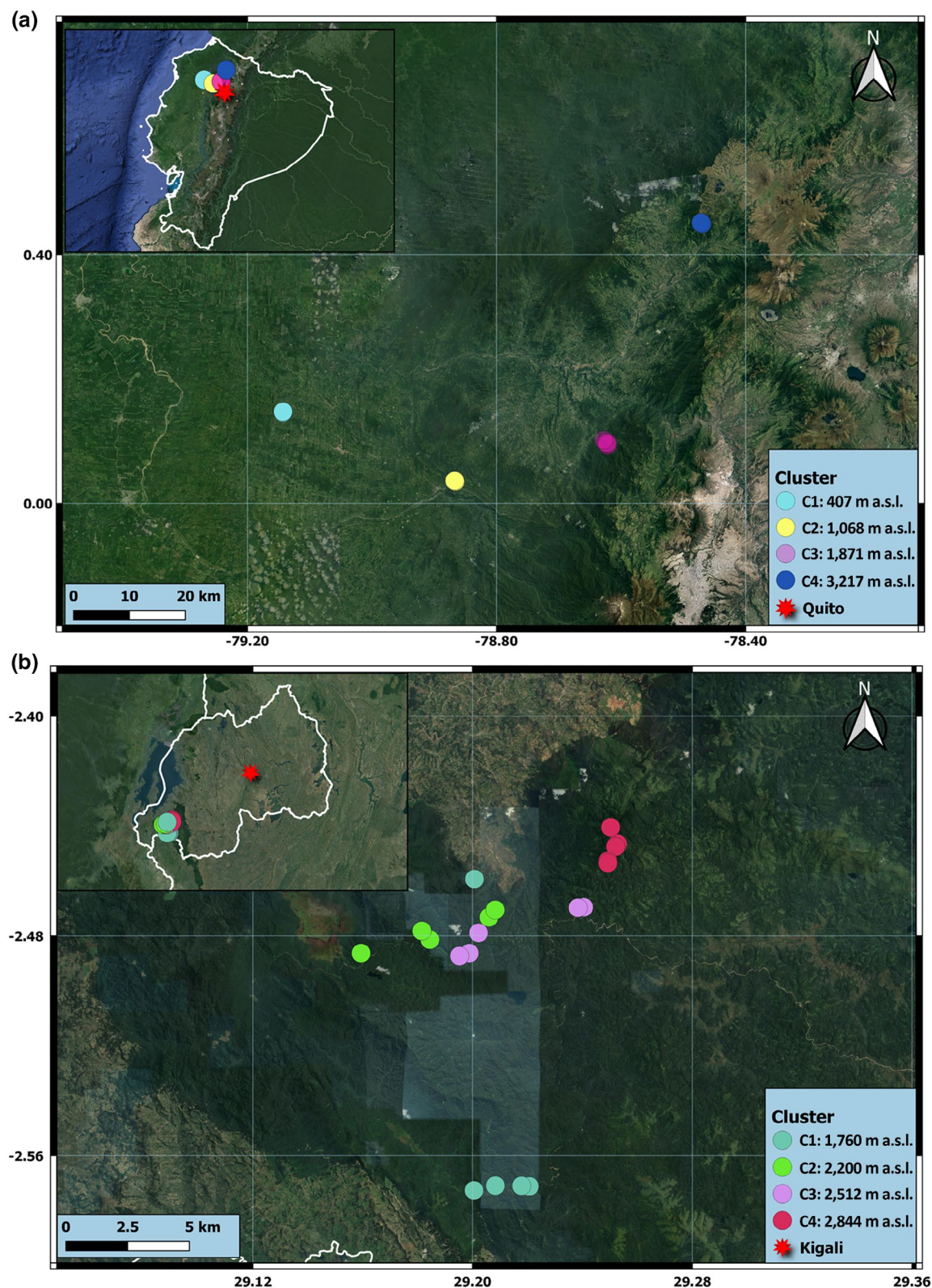


FIGURE 1 Plot locations on (a) the Ecuador transect (400–3,200 m a.s.l., 4–5 plots per elevational cluster) and (b) the Rwanda transect (1,600–3,000 m a.s.l., 5 plots per elevational cluster) projected on a cloud free Landsat composites false colour map

response variable at elevation 0 m, and β_{GE} quantified the extent to which the effect of elevation on the response variable differed between the two growth forms. Furthermore, we defined the

quantity $\delta = \beta_E + \beta_{GE}$ to capture the effect of elevation on the response variable for the non-reference growth form group (i.e. the trees).

TABLE 2 Fixed effect estimates in km a.s.l. (means) of the posterior distributions, and the quantile-based credible intervals for different canopy-level, and relative abundance and basal area response variables along the Ecuadorian and Rwandan transects

Response	Parameter	Estimate Ecuador	l-95% CI	u-95% CI	n	Estimate Rwanda	l-95% CI	u-95% CI	n
$\delta^{13}\text{C}$ (‰)					255				271
	β_I	-32.46	-34.73	-30.50		-30.64	-33.81	-27.58	
	β_E	0.55	-0.41	1.80		-0.14	-1.47	1.23	
	β_G	-1.00	-1.85	-0.14		-4.72	-7.21	-2.16	
	β_{GE}	0.80	0.36	1.23		2.32	1.17	3.43	
	δ	1.36	0.41	2.58		2.18	0.78	3.54	
N (g/m ²)					253				249
	β_I	1.01	-0.18	1.69		2.17	-3.14	4.26	
	β_E	0.18	-0.16	0.78		-0.03	-0.91	2.18	
	β_G	0.50	0.27	0.73		-1.08	-2.08	-0.08	
	β_{GE}	0.01	-0.11	0.12		0.63	0.18	1.07	
	δ	0.18	-0.15	0.78		0.60	-0.32	2.83	
P (g/m ²)					253				243
	β_I	0.062	-0.041	0.109		0.056	-0.251	0.185	
	β_E	0.008	-0.015	0.067		0.023	-0.033	0.155	
	β_G	0.012	-0.004	0.028		-0.013	-0.081	0.057	
	β_{GE}	0.009	0.002	0.017		0.010	-0.021	0.040	
	δ	0.018	-0.006	0.077		0.033	-0.024	0.163	
SLA (cm ² /g)					254				251
	β_I	251.97	213.85	292.81		101.34	28.66	163.62	
	β_E	-40.57	-58.18	-23.28		23.78	-4.49	57.24	
	β_G	-63.11	-100.81	-26.19		44.84	-21.11	110.90	
	β_{GE}	2.80	-12.97	18.10		-42.74	-72.62	-13.08	
	δ	-37.77	-52.72	-23.75		-18.96	-47.51	12.00	
$\delta^{18}\text{O}$ (‰)					249				257
	β_I	18.44	15.64	20.94		31.99	25.46	43.62	
	β_E	0.55	-0.75	1.99		-2.04	-7.07	0.78	
	β_G	2.25	1.26	3.23		-0.27	-4.84	4.20	
	β_{GE}	-0.69	-1.21	-0.18		-0.10	-2.08	1.93	
	δ	-0.15	-1.45	1.26		-2.14	-7.08	0.63	
C : N (-)					254				268
	β_I	14.35	9.97	18.77		15.18	7.94	24.01	
	β_E	1.89	-0.41	4.25		-0.47	-4.33	2.62	
	β_G	-2.00	-5.66	1.45		-5.39	-12.01	1.13	
	β_{GE}	4.39	2.31	6.62		4.70	1.75	7.68	
	δ	6.28	3.94	9.09		4.23	0.41	7.62	
C : P (-)					253				243
	β_I	264.58	153.53	383.55		396.38	225.94	561.08	
	β_E	53.14	-8.58	114.82		-45.25	-117.13	29.64	
	β_G	-30.93	-113.67	48.95		-43.00	-244.47	156.19	
	β_{GE}	58.84	16.21	100.03		70.37	-17.60	159.87	
	δ	111.98	56.22	165.27		25.11	-63.19	110.86	
N : P (-)					253				243

(Continues)

TABLE 2 (Continued)

Response	Parameter	Estimate	l-95% CI	u-95% CI	n	Estimate	l-95% CI	u-95% CI	n
		Ecuador				Rwanda			
	β_I	17.00	14.14	19.93		32.39	22.37	43.11	
	β_E	0.67	-0.68	2.02		-5.55	-10.25	-1.24	
	β_G	3.43	0.64	6.22		-9.36	-19.32	0.56	
	β_{GE}	-1.94	-3.28	-0.62		4.67	0.28	9.04	
	δ	-1.27	-2.41	-0.14		-0.88	-6.04	3.92	
rAbu (-)					17				20
	β_I	-0.76	-1.98	-0.30		-0.49	-2.82	1.74	
	β_E	-0.05	-0.53	0.45		-0.14	-0.90	0.62	
rBA (-)					17				20
	β_I	-4.54	-5.39	-3.80		-4.25	-5.67	-2.80	
	β_E	-0.05	-0.23	0.15		-0.01	-0.19	0.18	

Note: l-95% CI and u-95% CI are lower and upper limits of the 95% credible interval, respectively; β_I , β_E , β_{GE} , δ ($\beta_E + \beta_{GE}$) are regression coefficients for the intercept, the elevation effect for lianas, the growth form, the interaction between growth forms (lianas and trees) and the elevation effect for trees, respectively; $\delta^{13}\text{C}$, N, P, SLA, $\delta^{18}\text{O}$, C : N, C : P, N : P, rAbu, rBA are response variables, respectively, for foliar carbon stable isotopic composition, foliar nitrogen, foliar phosphorus, specific leaf area, foliar oxygen stable isotopic composition, carbon-to-nitrogen ratio, carbon-to-phosphorus ratio, nitrogen-to-phosphorus ratio, relative abundance for lianas and relative basal area for lianas. For estimated parameters of all models, bulk and tails ESS > 100 $\times$ 5 and 1 $\leq$ Rhat < 1.1. ESS is the effective sample size and Rhat is the potential scale reduction on split chains. Parameter estimates whose CIs do not contain zero are indicated in bold.

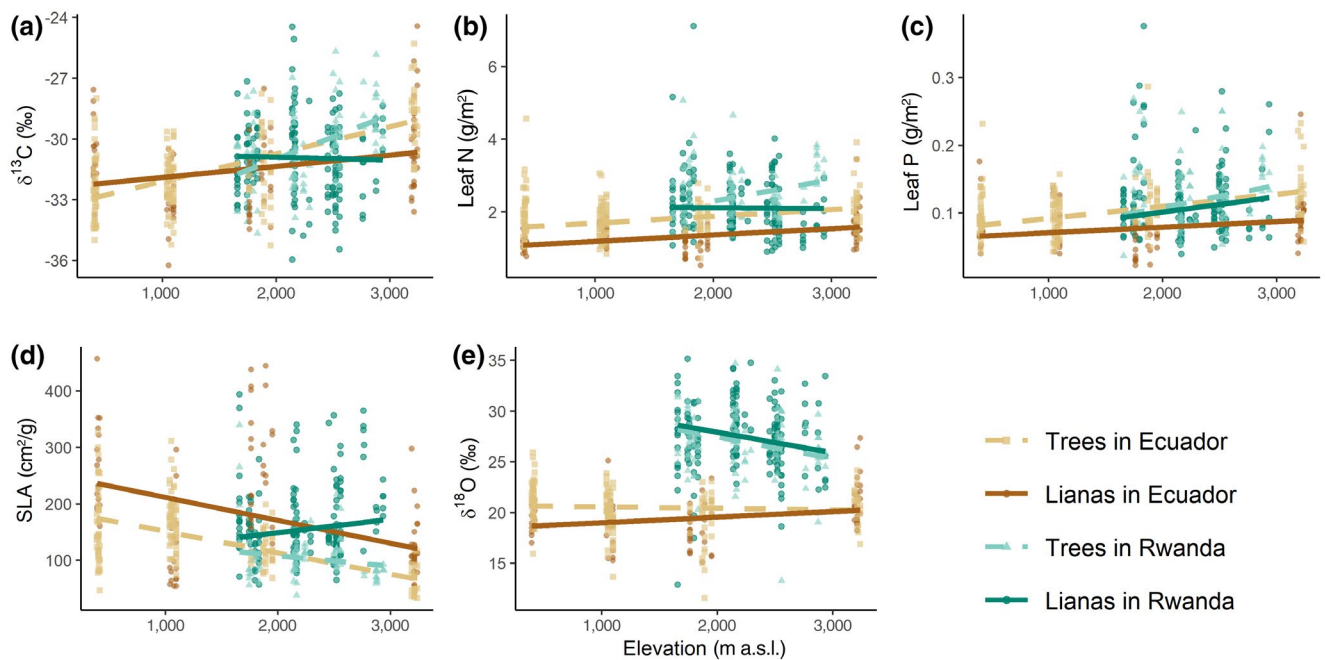


FIGURE 2 Shifts in foliar carbon stable isotopic composition ($\delta^{13}\text{C}$; a), foliar nitrogen (N; b), foliar phosphorus (P; c), specific leaf area (SLA; d) and foliar oxygen stable isotopic composition ($\delta^{18}\text{O}$; e) with elevation for lianas (solid lines) and trees (dashed lines) along transects in Ecuador (brown palette, 400–3,200 m a.s.l.) and Rwanda (green palette, 1,600–3,000 m a.s.l.). Each line denotes the posterior mean of the fixed effect elevation. Dots represent individual data points

We used Bayesian robust correlations to explore relationships between foliar $\delta^{13}\text{C}$ and other foliar traits, including foliar $\delta^{18}\text{O}$, foliar nutrient concentrations (N_{area} , P_{area}), the mass stoichiometric ratios (C : N, C : P, N : P) as well as SLA. For each bivariate correlation, we set up a model with a bivariate response variable and without predictors (intercept only model). For each model, we estimated the

covariance and residual variance to obtain the correlation value and its standard deviation.

We explored the change in liana relative abundance and BA with elevation by setting up Bayesian LMMs with elevation as a fixed effect and elevational cluster as random intercept effect.

We conducted all analyses using R 4.0.3 (R core Team, 2020). We used the Stan computational framework (Stan Development Team, 2014) from the package 'brms' (Bürkner, 2017) to fit all the Bayesian models. Methodological details on the Bayesian model fitting and diagnosis can be found in Supporting Information Appendix S1.

We drew reasonable inferences from our Bayesian (generalized) LMMs (Table 2) after estimating the density curves of posterior distributions of population-level regression parameters (Supporting Information Figure S1) and ensuring that the models captured the important features of the data (Supporting Information Figure S2). The credible intervals (CIs) reported alongside the estimated fixed effects (Table 2) show the central portions of the posterior distributions that contain 95% of the values. The effect estimates were considered statistically meaningful if the 95% CIs did not contain zero.

3 | RESULTS

3.1 | Differences in the responses of $\delta^{13}\text{C}$ and other foliar traits to elevation between lianas and trees

Given the model and the data, we were able to assert with confidence that there is a meaningful difference in the foliar $\delta^{13}\text{C}$ response to elevation between liana and tree species on both transects (Table 2). Foliar $\delta^{13}\text{C}$ meaningfully increased with elevation for tree species while it did not meaningfully change for lianas on both transects (Figure 2a). Additionally, we found evidence of different responses to increasing elevation between lianas and trees in terms of foliar N_{area} (in Rwanda), foliar P_{area} (in Ecuador), SLA (in Rwanda) and foliar $\delta^{18}\text{O}$ (in Ecuador) (Table 2). Despite these findings, we could not confidently ascertain that foliar N_{area} and P_{area} for both growth forms and on both transects meaningfully changed with increasing elevation (Figure 2b–c). We noted a meaningful decrease in SLA with increasing elevation for both growth forms on the Ecuadorian transect. While we detected a marginal increase in

SLA for lianas in Rwanda, we were less confident about a meaningful decrease in SLA with elevation for trees (Figure 2d). We found a marginal decrease in foliar $\delta^{18}\text{O}$ with elevation for both growth forms in Rwanda whereas we were not confident about a meaningful change in foliar $\delta^{18}\text{O}$ with increasing elevation for both growth forms in Ecuador (Figure 2e).

3.2 | Differences in the responses of foliar elemental ratios to elevation between lianas and trees

Given the model and the data we could state with confidence that lianas and trees responded differently to increasing elevation in terms of foliar C : N (on both transects), foliar C : P (in Ecuador) and foliar N : P (on both transects) (Table 2). Foliar C : N meaningfully increased for trees with elevation on both transects, marginally increased for lianas in Ecuador, and remained relatively constant for lianas in Rwanda (Figure 3a). Foliar C : P meaningfully increased for trees and marginally increased for lianas with increasing elevation on the Ecuadorian transect. In Rwanda, foliar C : P marginally decreased with elevation for lianas while it remained relatively constant for trees (Figure 3b). Foliar N : P meaningfully decreased with elevation for trees in Ecuador and lianas in Rwanda while elevation exerted no meaningful effect on foliar N : P for lianas in Ecuador and trees in Rwanda (Figure 3c).

3.3 | Changes in liana relative abundance and basal area with elevation

Along both transects, and for both growth forms, stem density peaked at the third elevational cluster (c. 2,000 m in Ecuador and c. 2,500 m in Rwanda). Consequently, liana relative abundance was highest at these elevational clusters (44% and 55% on the Ecuadorian and Rwandan transects, respectively; Table 1). However, liana relative abundance decreased to its lowest level at the highest elevational cluster, c. 3,000 m on both transects (23% and 9% on

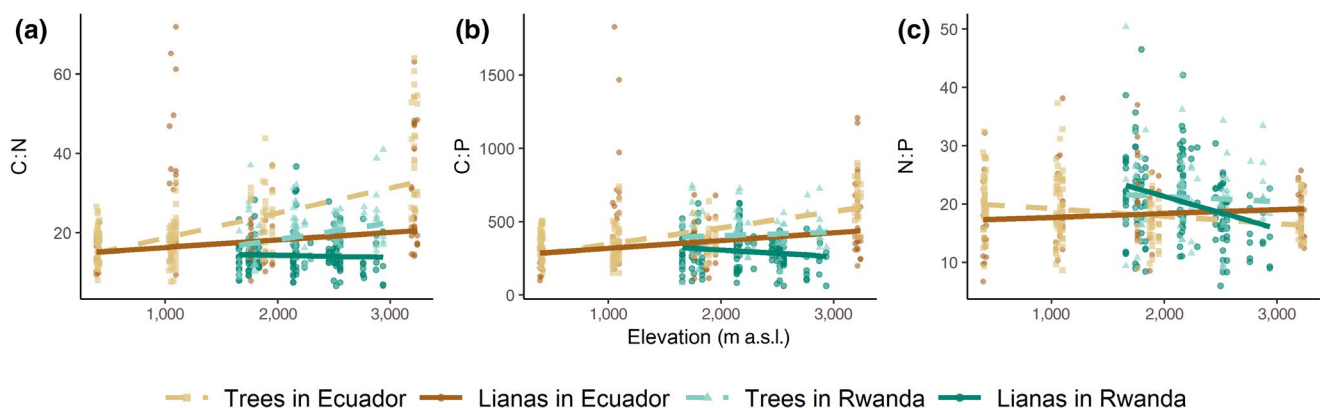


FIGURE 3 Shifts in foliar C : N (a), foliar C : P (b) and foliar N : P (c) with elevation for lianas (solid lines) and trees (dashed lines) along transects in Ecuador (brown palette, 400–3,200 m a.s.l.) and Rwanda (green palette, 1,600–3,000 m a.s.l.). Each line denotes the posterior mean of the fixed effect elevation. Dots represent individual data points

the Ecuadorian and Rwandan transects, respectively; Table 1). Liana relative BA followed the same pattern as that described for liana relative abundance on both transects (Table 1). The outputs of the statistical models revealed that the rates of changes in liana relative abundance and basal area with increasing elevation were not statistically robust on both transects (Table 2, Figure 4a–b).

4 | DISCUSSION

4.1 | Differences in the responses of $\delta^{13}\text{C}$ and other foliar traits to elevation between lianas and trees

This study reveals interesting differences in foliar $\delta^{13}\text{C}$ responses to increasing elevation between liana and tree species. Foliar $\delta^{13}\text{C}$ meaningfully increased with elevation for trees at the rate of 1.4 and 2.2‰ per km on the Ecuadorian and Rwandan transects, respectively. The rate of increase in leaf $\delta^{13}\text{C}$ that we report for trees in Rwanda is particularly striking as it is almost twice as high as what is globally reported for non-water stressed regions (c. 1‰ per km, 50–4,740 m a.s.l.; Körner et al., 1988). For lianas, however, the rates of change in foliar $\delta^{13}\text{C}$ with elevation in Ecuador (0.6‰ per km) and Rwanda (–0.1‰ per km) were not statistically meaningful. The question therefore arises as to why co-existing species of these two growth forms show different foliar $\delta^{13}\text{C}$ responses to elevation on these transects. Increasing foliar $\delta^{13}\text{C}$ with elevation is a usual trend in humid regions, and has been documented in several studies (Hultine & Marshall, 2000; Körner et al., 1988). Although this can be explained by several factors, the two consistent patterns associated with increasing elevation on a global scale are a decrease in partial pressures of both CO_2 and O_2 , and in temperature (Wang et al., 2017). The decrease in O_2 partial pressure and temperature supposedly promotes a decline in c_i/c_a by increasing the affinity of Rubisco for CO_2 , and thus increasing its

carboxylation efficiency (Wang et al., 2017). Declining temperature also increases the viscosity of water (η), which may hinder the flux of water in plants, resulting in partial stomatal closure and lower c_i/c_a (Roderick & Berry, 2001). The direct implication of a decline in c_i/c_a is that discrimination against $^{13}\text{C}\text{-CO}_2$ decreases, $\delta^{13}\text{C}$ values become less negative, and long-term iWUE increases (Farquhar & Richards, 1984).

Because of the large amount of N invested in the Rubisco enzyme, foliar N_{area} is often positively correlated to photosynthetic rates (Evans, 1989). High foliar N_{area} is expected to result in higher A_n , leading to a higher drawdown of CO_2 , lower level of discrimination against $^{13}\text{C}\text{-CO}_2$ and less negative $\delta^{13}\text{C}$ values (Friend et al., 1989). Although decreasing foliar N_{area} with elevation has been reported (Zhang et al., 2005), the common trend is that foliar N_{area} increases with elevation (Sparks & Ehleringer, 1997). Along our transects, foliar N_{area} for both trees and lianas did not meaningfully change with elevation. The same observation was valid for foliar P_{area} , another nutrient involved in photosynthetic activity. Therefore, we are inclined to believe that the meaningful increase of $\delta^{13}\text{C}$ with elevation observed for tree species along these transects is not attributable to a higher A_n . These results suggest that the observed divergent pattern of $\delta^{13}\text{C}$ between these two growth forms along these two elevational gradients may not be explained by potential differences in A_n .

SLA decreased with elevation for both growth forms in Ecuador. This trend is in agreement with many previous studies (Hultine & Marshall, 2000; Kao & Chang, 2001). It implies that leaves of both growth forms become thicker with elevation on this transect. In Rwanda, SLA showed a decreasing trend for trees (although not meaningful) and a marginal increase for lianas with elevation. Decreasing SLA with elevation has often been attributed to the lower temperatures at high altitudes (Kao & Chang, 2001). It is believed that plants at high elevation with thicker leaves might be impacted by a longer internal diffusional pathway for CO_2

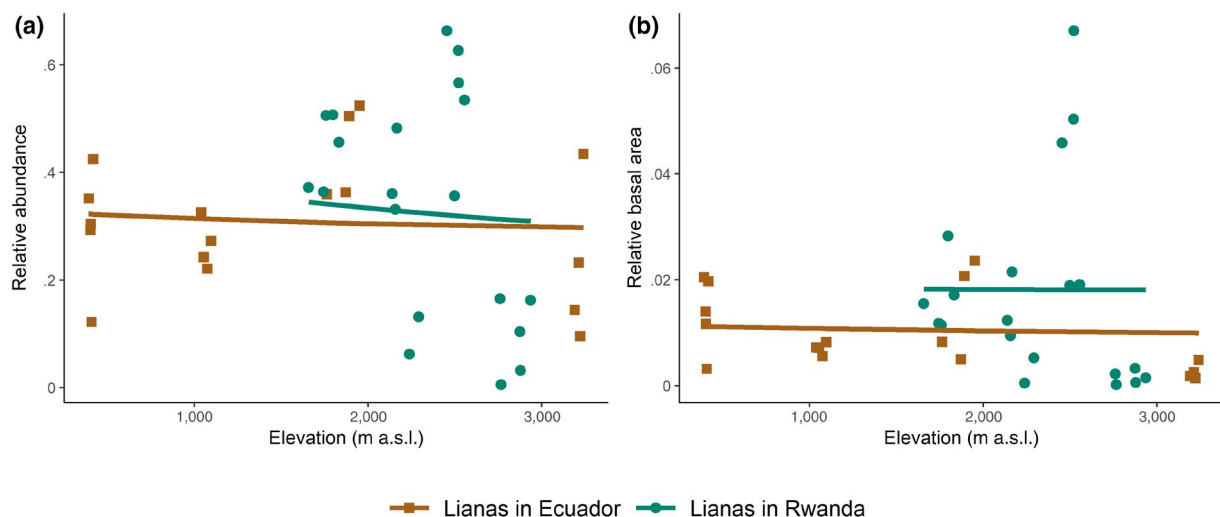


FIGURE 4 Shifts in liana relative abundance (a) and relative basal area (b) with elevation along transects in Ecuador (brown palette, 400–3,200 m a.s.l.) and Rwanda (green palette, 1,600–3,000 m a.s.l.). Each line denotes the posterior mean of the fixed effect elevation. Dots represent individual data points

from the atmosphere to the intercellular air space, leading to a potential decrease in c_i/c_a (Vitousek et al., 1990). However, Wang et al. (2017) argued that low air pressure at high altitudes should allow CO_2 to diffuse more easily through the stomata, counteracting the possible effect resulting from a lengthened CO_2 diffusional pathway. The increase in leaf thickness (and resulting lengthened CO_2 diffusional pathway) is therefore likely not a plausible explanation of the contrasting $\delta^{13}\text{C}$ pattern between lianas and trees on both transects. Elevational decrease in SLA has sometimes been associated with a decrease in g_m , which may induce a decrease in the chloroplast (c_c) to atmospheric CO_2 partial pressure ratio (c_c/c_a), leading to less negative $\delta^{13}\text{C}$ values (Qiuhong et al., 2013). This might give a possible explanation of the contrasting pattern of $\delta^{13}\text{C}$ between lianas and trees with increasing elevation on the Rwandan transect (Figure 2a & d).

Foliar $\delta^{18}\text{O}$ was used to get an insight into the difference in water loss (and CO_2 supply) through stomata between trees and lianas with increasing elevation. The lack of a clear positive relationship between this parameter and elevation for both lianas and trees along both transects suggests the absence of a greater evaporative enrichment (i.e. the presence of ^{18}O enriched leaf water) caused by either higher leaf-to-air vapour pressure deficit (VPD) or tighter stomatal control (Scheidegger et al., 2000). Similarly, the absence of a strong positive correlation between $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$ for both growth forms (Supporting Information Table S1) may tend to suggest that ^{13}C - CO_2 discrimination by Rubisco is not driven by variation in g_s along these transects (Barbour, 2007). However, caution has been raised over the use of the $\delta^{13}\text{C}$ - $\delta^{18}\text{O}$ dual-isotope approach in study locations characterized by high relative humidity (RH) levels (Roden & Farquhar, 2012). For example, Bauters et al. (2020) have recently shown that $\delta^{18}\text{O}$ was insensitive to g_s for central Africa rain forests, due to a masking effect of the persistent high RH. Therefore, $\delta^{18}\text{O}$ might be an insensitive proxy for g_s in the tropics, and should be interpreted with great care in this study.

Lianas are believed to have a higher g_s on average than trees (see Wyka et al., 2013). However, lower g_s in lianas than trees (Johnson et al., 2013) or no significant difference in g_s between these two growth forms (see Mello et al., 2020) have also been reported. In a recent study, van der Sande et al. (2019) found that g_s was negatively correlated with iWUE for trees, but they found no correlation between g_s and iWUE, and a weak positive correlation between A_n and iWUE for lianas. They concluded that the iWUE of trees is mostly driven by variation in g_s whereas iWUE of lianas is mainly driven by variation in A_n . Building on their findings, we believe that, along these two transects, g_s likely declined for trees with increasing elevation leading to increasing iWUE. As elevation increases, the diffusion of water from the leaves into the air in the form of vapour (transpiration, E) increases due to reduced atmospheric pressure (Gale, 2004), and water absorption by roots decreases due to lower soil temperatures (see Göbel et al., 2019). Additionally, decreasing temperatures induce an increase in the η that slows down the transport of water from the soil to the evaporative sites in leaves

(Roderick & Berry, 2001). All these effects induce drought stress, prompting trees to reduce their g_s as a water conservation strategy.

Compared to tree species, evidence suggests that liana species are less affected by seasonal drought (Cai et al., 2009). Lianas seem to exhibit a water-use advantage over trees in seasonal tropical forests due to a tighter diurnal stomatal control (Chen et al., 2015). Furthermore, lianas have an efficient shallow root system (De Duerwaerder et al., 2018). They are also known to possess a high stem hydraulic conductance (van der Sande et al., 2019), and maintain leaf water potential within a safe hydraulic range in order to avoid xylem dysfunction (Campanello et al., 2016). These characteristics make them very efficient in terms of water acquisition and transport even during the dry season (see Schnitzer, 2018). For instance, it has been shown that lianas can maintain the same rate of photosynthesis (Zotz & Winter, 1996), and a similar g_s (see Castellanos, 1991) during the dry and wet seasons. This implies that certain liana species may be able to maintain a relatively constant iWUE during the dry and wet seasons. Therefore, we believe that there was less need for lianas to significantly minimize their E and reduce their g_s to withstand drought stress along these two elevational transects.

The aforementioned considerations on the observed foliar $\delta^{13}\text{C}$ elevational patterns for lianas and trees are in line with the theory proposed by Prentice et al. (2014). This theory posits that the optimal c_i/c_a should depend on minimizing the summed cost function of maintaining E and A_n . The cost function of maintaining A_n can be approximated by the ratio of total respiration to carboxylation capacity (V_{cmax}). The cost function for maintaining E , for its part, depends on a certain number of parameters including, among others, η and the ratio of sapwood density to sapwood permeability (ρ_s/k_s). The parameter η changes as a function of temperature, and ρ_s and k_s vary among species. Compared to trees, lianas typically have a lower ρ_s and a higher k_s (Apgaua et al., 2017). Hence, ρ_s/k_s is expected to be smaller for lianas than for trees. Therefore, the maintenance cost of E should increase at a slower rate with decreasing temperature for lianas than for trees. The overall implication, according to the theory, is the optimal c_i/c_a decreases less with decreasing temperature (and hence elevation) for lianas compared to trees.

4.2 | Differences in the responses of foliar elemental ratios to elevation between lianas and trees

Alongside a relatively constant $\delta^{13}\text{C}$, lianas maintained their C : N ratios relatively constant along both transects. This indicates that lianas may keep a relatively constant N use efficiency with increasing elevation. Tree species, on the contrary, showed an increase in C : N ratios with elevation on both transects, suggesting a probable progressive increase of their N use efficiency (Agren, 2004). We found no evidence for an increased N or P limitation with elevation for lianas on the Ecuadorian transect as both C : N and N : P ratios remained relatively constant. Therefore, the abundance

and BA of lianas are unlikely to be limited by nutrients along the Ecuadorian transect in line with a recent nutrient-addition experiment in Panama (Schnitzer et al., 2020). In Rwanda, the declining N : P ratio with elevation for lianas suggests a progressive shift from P-to-N limitation. The increase in C : N (both transects) and decrease in N : P (Ecuador only) ratios for trees indicate that this growth form was progressively becoming N limited with increasing elevation. This is in accordance with other studies along elevational gradients in the tropics (Bauters et al., 2017; Fisher et al., 2013). These observations indicate that the shift from P-to-N limitation while transitioning from lowland to montane tropical forests (Townsend et al., 2008), as commonly reported, may vary among sites and growth forms of woody species.

4.3 | Changes in liana relative abundance and basal area with elevation

Liana abundance generally decreases with increasing elevation (Alves et al., 2011; Fadrique & Homeier, 2016). The large vessels of lianas and their acquisitive vascular system are very susceptible to the cold temperatures associated with high elevations (Jiménez-Castillo et al., 2007), thus explaining low liana abundance at high elevations. Along our transects, liana relative abundance and BA did not exhibit strong decreasing trends as we could have expected. Consequently, the contributions of lianas to the total abundance and BA of woody species did not change too much along most parts of both transects. This suggests that lianas could maintain their competitive ability along these transects. Despite these overall trends, we noticed that the relative abundance and BA of lianas peaked at mid-elevations (probable optimum growth zones), and dropped to their lowest levels at the highest elevations on both transects. This observation may suggest that lianas might be more competitive at mid-elevations (c. 2,000 m in Ecuador and c. 2,500 m in Rwanda), and less at the highest elevations (c. 3,000 m on both transects) along these transects. The liana species present at the highest elevation must have developed some mechanisms allowing them to overcome the risk of cold-induced embolism (see Schnitzer, 2018) while maintaining relatively the same iWUE as their counterpart species present at lower elevations.

We speculate that the abundance and BA of lianas and trees along these elevational transects could be determined by two major factors, namely the level of drought stress and that of cold stress. As drought stress becomes more stringent along both transects, tree species seem to feel that effect more severely than lianas, prompting them to increase their iWUE by tightly regulating their g_s . If drought stress increases further along these transects, trees might be at higher risk of mortality, especially by carbon starvation (O'Brien et al., 2014) as excessive stomatal closure can lead to reduced photosynthesis. On the contrary, lianas might still have enough room to adjust their iWUE to cope with higher levels of drought stress. However, this advantage at the highest elevations will only be true for the limited number of liana species that would have been able to

overcome the risk of embolism in their xylem vessels associated with low temperatures.

5 | CONCLUSION

Lianas exhibited stronger physiological damping of environmental forcing on $\delta^{13}\text{C}$ as compared to trees along two elevation transects in Ecuador and Rwanda. The $\delta^{13}\text{C}$ signature of liana leaves did not change consistently with elevation, while tree leaf $\delta^{13}\text{C}$ increased with elevation. Additionally, no meaningful elevational trend in C : N ratio was detected for lianas while an increase was observed for trees. This suggests that lianas may operate at relatively constant iWUE and N use efficiency along both transects. Trees, on the contrary, increased their iWUE, most likely by tightly controlling their stomatal conductance. At the same time, they appeared to increase their N use efficiency. This suggests that tropical liana and tree communities adopt divergent functional strategies with respect to water and nutrient use on these two elevation transects. If the temperature decreases further along these transects, tree communities might be more restricted by water and nutrient availability whereas liana communities might suffer more from cold stress (embolism). However, more work comparing the eco-physiological responses to elevation between lianas and trees across tropical regions is needed to confirm our findings, and validate the proposed mechanisms.

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CONFLICTS OF INTEREST

The corresponding author confirms on behalf of all authors that there are no conflicts of interest.

AUTHOR CONTRIBUTIONS

H.V., M.B., P.B. and S.B. planned and designed the research. M.D., C.M., L.V. and J.P. conducted plant inventories and collected data in Ecuador. B.S., L.N.C. and M.B. conducted plant inventories and collected data in Rwanda. F.M.M. analysed the data and wrote the manuscript. F.M., L.A.C., H.P.T.D. and C.A.L. revised the paper. All authors agreed on the final version of the manuscript.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in the Dryad Digital Repository: <https://doi.org/10.5061/dryad.tqjq2bw0d> (Mumbanza et al., 2022).

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BIOSKETCHES

CAVElab is studying vegetation dynamics, carbon and water cycling, with a strong focus on tropical forest ecosystems.

ISOFYs has a general interest in the usage of stable isotopes for disentangling the biogeochemical cycles of natural and human ecosystems, with a strong focus on global change related research.

SUPPORTING INFORMATION

Additional Supporting Information may be found online in the Supporting Information section.

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